

Inference and effect size

Activity

Work on the activity (handout), then we will discuss as a class.

Activity

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.73743	0.08499	20.44	<2e-16	***
WBC	-0.36085	0.01243	-29.03	<2e-16	***

Calculate a 99% confidence interval for the change in the log odds of dengue associated with a one-unit increase in WBC.

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How could we calculate a 99% confidence interval for the change in the *odds* of dengue associated with a one-unit increase in WBC?

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99% confidence interval in terms of odds: (0.675, 0.720)

How do I interpret this confidence interval?

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99% confidence interval in terms of odds: (0.675, 0.720)

Your friend, seeing your confidence interval from question 2, states that there is a 1% chance that the true change in odds associated with a one-unit increase in WBC is less than 0.675 or greater than 0.720. Is your friend correct?

Types of inference questions

- + Is there a relationship between these variables?
 - + Hypothesis testing
- + What is the size of relationship between these variables?
 - + Confidence intervals

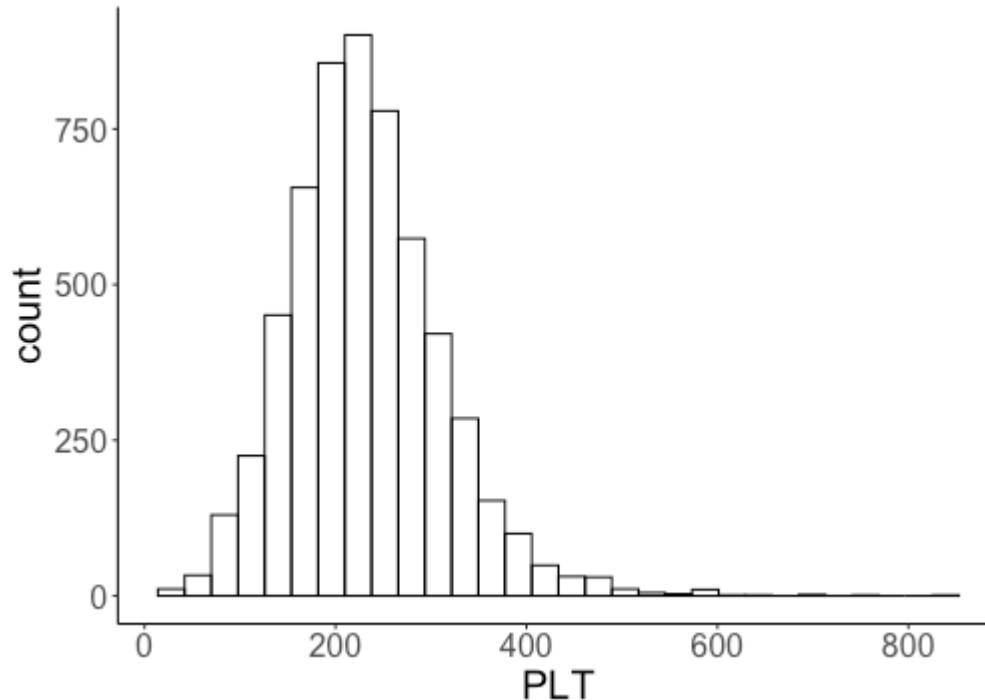
Today: thinking about effect size

Thinking about effect size

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.2525593	0.1548038	8.091	5.9e-16	***
Age	0.1383186	0.0099763	13.865	< 2e-16	***
WBC	-0.2523294	0.0135371	-18.640	< 2e-16	***
PLT	-0.0060276	0.0006113	-9.860	< 2e-16	***

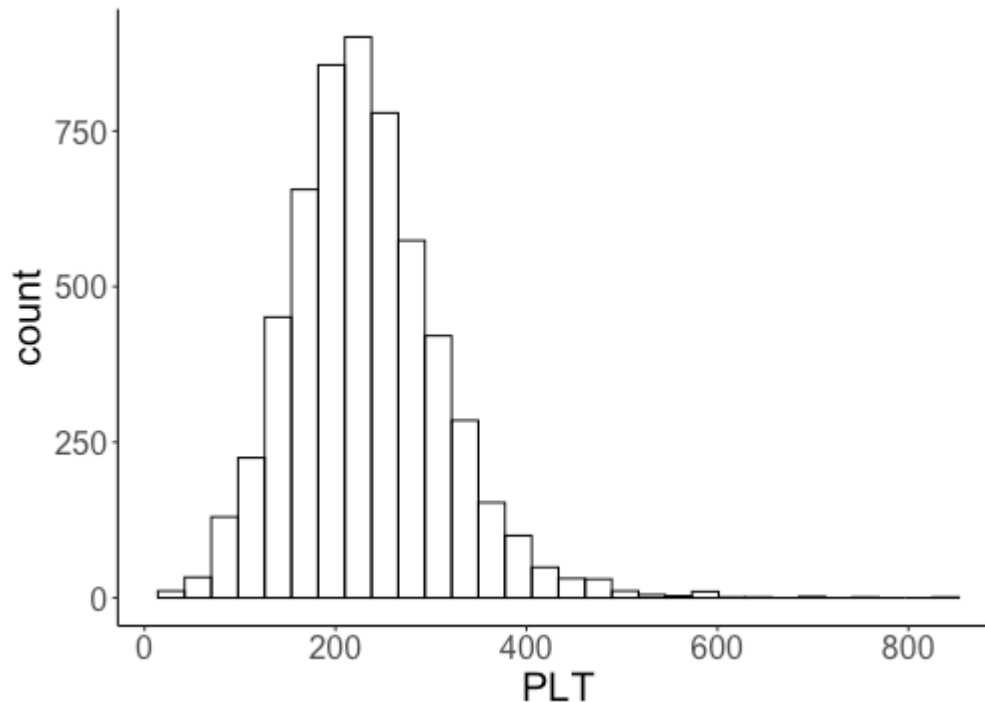
Your friend claims that the coefficient on platelet count (PLT) is very close to 0, so platelet count is not a meaningful variable to include in the model. Is your friend correct?

Thinking about effect size



- + The coefficient on PLT (-0.006) is interpreted in terms of a one-unit increase
- + The scale of PLT ranges from 0 to 800 -- a one-unit increase is very small!

Thinking about effect size



If a one-*unit* increase has different meaning across different variables, how else could we think about the effect size of a variable?

Effects in terms of standard deviations

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.2525593	0.1548038	8.091	5.9e-16	***
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WBC	-0.2523294	0.0135371	-18.640	< 2e-16	***
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The *standard deviation* of PLT in the data is 80.02.

What is the change in log-odds associated with a one *standard deviation* in PLT, holding the other variables constant?

Effects in terms of standard deviations

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The *standard deviation* of PLT in the data is 80.02.

Holding Age and WBC constant, a one *standard deviation* increase in PLT is associated with:

- + A decrease of -0.482 in the log-odds of dengue
- + A change in the *odds* by a factor of $e^{-0.482} = 0.617$ (i.e., a decrease of almost 40%)!

Effects in terms of standard deviations

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The *standard deviation* of PLT in the data is 80.02.

The estimated probability of dengue at the mean values of Age, WBC, and PLT is about 0.2. If we fix Age and WBC:

- + Probability when $PLT = \text{mean} + 1 \text{ sd}$ is 0.135
- + Probability when $PLT = \text{mean} - 1 \text{ sd}$ is 0.292

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	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.507329	0.311939	-8.038	9.14e-16	***
loss	-0.846499	0.238335	-3.552	0.000383	***
smoke100	-0.137151	0.035816	-3.829	0.000128	***
hlthplan	0.613613	0.051233	11.977	< 2e-16	***
age	-0.012148	0.001043	-11.652	< 2e-16	***
height	0.056845	0.004520	12.577	< 2e-16	***

The standard deviation of the `loss` variable is 0.074.

How do the odds of exercise change with a one standard deviation increase in `loss`, holding the other variables constant?

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loss	-0.846499	0.238335	-3.552	0.000383	***
...	...	...	...	...	

The standard deviation of the `loss` variable is 0.074.

Want to test whether a one-standard deviation increase in `loss` is associated with a decrease in the odds of at least 5% (holding other variables constant).